



DECK DEPARTMENT TRAINING

Mooring Operations

Mooring systems, safe practice, winches and windlasses, ropes and wires, and their maintenance.

OCIMF MEG

Company Manual

Maker instructions

Terminal rules

What we will cover

01

The Mooring System

Purpose, forces, arrangements and configurations

02

Safe Mooring Practice

PPE, snap-back, stoppers, capstans and fire wires

03

Organising the Operation

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Winches and Windlasses

Types, brakes, anchoring and recovery

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Synthetic Fibre Ropes

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Periodic checks, lubrication and brake tests



SECTION 01

The Mooring System

What the mooring system is for, the forces it has to hold, and the arrangements used to hold them.

01

The purpose of mooring

The mooring system exists to position the ship correctly and to keep her there while work is carried out.



Position the ship

Hold the ship in the correct position alongside during cargo loading and discharging operations.



Shift the ship

Mooring lines can also be used to move the ship along the quay when required.



Hold her for the stay

The lines keep the ship secure for the whole of the port stay, not only during cargo work.

What the system must hold against

The mooring system must protect the ship's position against the effect of one or more of the following, acting together.



Wind

The dominant force on a well-sited berth, and it grows with height above the water.



Current

Acts on the immersed hull; its effect depends heavily on under-keel clearance.



Passing ships

Interaction from ships passing close by produces surge and shock loads in the lines.



Ice

Ice loads add a further horizontal force on the hull at the berth.



Waves and swell

Waves and swell make the ship work against her lines, especially at exposed berths.



Changing ship condition

Freeboard, under-keel clearance, trim and list all change as cargo and ballast are worked.

Typical tanker terminal arrangement

A typical tanker berth combines a central loading platform with breasting and mooring dolphins.



Loading platform

The central structure carrying the manifold connection, positioned against the ship's cargo manifold.



Breasting dolphins

The fendered structures that take the ship's side and absorb the berthing energy.



Mooring dolphins

Separate outlying structures, fitted with hooks, to which the breast and spring lines are led.



Breast lines

Led as nearly perpendicular to the ship's side as possible; they hold the ship in against the fenders.



Spring lines

Led nearly parallel to the ship's side; they restrain fore-and-aft movement along the berth.

Head and stern lines against breast lines

The two groups of lines do different work, and they are not interchangeable.



Head and stern lines

Led well forward and well aft

- Run at a shallow angle away from the ship, towards the shore ahead and astern
- Do not hold the ship in against the berth
- Relatively less effective than breast lines for keeping the ship alongside
- Contribute mainly to longitudinal restraint



Breast lines

Led perpendicular to the side

- Led as close to perpendicular to the ship's side as the berth allows
- Hold the ship in against the fenders
- The most effective lines for controlling off-berth movement
- Should be paired forward and aft to balance the ship

The mooring arrangement plan

Every ship carries a mooring arrangement plan; the example in this course is for a 158,000 dwt tanker.



General arrangement of the deck

A plan view showing forecastle, main deck, manifold and poop with every fitting in place.



Every fairlead and bitt

The position, type and lead direction of each fairlead, pedestal roller, bitt and chock is shown.



Winches and drums

Winch positions, drum arrangements and the lines led from each drum are identified.



Fitting schedules

Tables list the safe working load of each fitting, so no line is led to a fitting that cannot take it.



Distances marked

Distances between fittings are given, so the lead angles at any berth can be worked out in advance.



Use the plan, not memory

The plan is the only reliable record of what each fitting will hold. Before mooring at an unfamiliar berth, check the intended lead against the plan.

The five mooring configurations covered

Beyond the conventional berth, four further arrangements are used depending on the trade and the location.



Alongside berth

Conventional quay or dolphin berth with head, stern, breast and spring lines.



Anchorage

The ship holds her position on her own ground tackle, without shore structures.



SPM

Single point mooring: the ship is secured to one buoy and swings freely.



STS

Ship-to-ship: two vessels of different size secured together for cargo transfer.



MBM

Multi buoy mooring: two anchors from forward, buoys aft.

Anchoring as a mooring method

The ship's mooring equipment includes her ground tackle, and anchoring is a mooring operation in its own right.



What it does

- Stabilises the ship's position away from land
- Does not require the ship's propulsion power to be used to hold position
- Uses only the ship's own equipment, with no shore assistance



What it depends on

- The holding power of the anchor in the seabed
- The weight and catenary of the chain paid out
- The condition of the windlass, brake and chain stopper
- The care with which the anchor is let go and recovered

Single point mooring (SPM)

At an SPM the ship is secured to a buoy by one or two special mooring lines, and the entire load is carried by them.



Special mooring lines

The ship is tied to the buoy using dedicated rope or ropes supplied for the purpose. The whole of the mooring load is on them.



SPM stopper

The mooring lines are secured on board at the SPM stopper on the forecastle, not on ordinary bitts.



Anchor swivel

A swivel in the buoy anchoring arrangement allows the ship to turn freely under the effect of wind and current.



Floating hose

Cargo is transferred through a floating hose taken to the tanker manifold, separately from the mooring lines.



Buoy base

The buoy is held to the seabed by its own anchoring system, which carries the load transmitted from the ship.

Ship-to-ship transfer

An STS transfer requires two ships of different size to be secured together, away from land, for cargo transfer.



Two ships, different sizes

Typically a larger vessel and a smaller one, secured side by side on fenders.



Under way or moored

The operation can also be performed while one of the ships is moored, or where more than two vessels are involved.



Fenders take the load

Contact between the hulls is taken entirely on the fenders, so their placement is part of the plan.

STS manoeuvring — points never to overlook

Each Master will manoeuvre according to his own preference, but the following points apply to every STS operation.

✓ DO

- Both Masters must be prepared to abort the mooring operation at any time
- Both ships have a duty to keep a proper lookout at all times
- Take the wind and sea very close to ahead, or on the bow, during the manoeuvre
- Apply an alternative approach where local conditions or experience require it
- Predict the results of interaction between two large ships before closing

⊘ DON'T

- Do not use a large approach angle when closing the other ship
- Do not plan a manoeuvre that requires the engine to be worked astern during the approach
- Do not continue an approach that both Masters are not content with
- Do not rely on a single method regardless of local circumstances
- Do not underestimate hydrodynamic interaction between the two hulls

The common STS approach method

The most widely used approach, with the constant-course ship taking the load off the manoeuvre.

1

Constant course and speed

The ship to be berthed alongside maintains a steady course and a fixed, slow speed throughout.

2

Approach from astern

The manoeuvring ship approaches from astern, on the side on which she will berth. Approach distances of the order of 1,000 m astern and 500 m off are used.

3

Parallel up at 50 to 100 m

When the transverse distance falls to approximately 50 to 100 metres, the approaching ship adjusts course to run parallel to the other ship.

4

Close on small orders

Speed is matched to the other ship and contact with the fenders is made using small rudder and engine orders only.

5

No astern movement

The manoeuvre should not require the engine to be worked astern, and interaction effects must be anticipated throughout.

STS approach — the distances that matter

The figures used in the course diagram for the approach of the manoeuvring ship to the constant-course ship.

1,000 m

Distance astern at which the approach is commenced

500 m

Lateral offset held during the run up from astern

50 – 100 m

Transverse distance at which the ship parallels up



Approach from the berthing side — The manoeuvring ship comes up from astern, on the side on which she is to berth.



Then small orders only — From the parallel position, speed is matched and the fenders are met using small rudder and engine orders.

Multi buoy mooring (MBM)

MBM is generally used where weather and sea conditions are between good and moderate.



Forward

Two anchors

- The ship anchors with two anchors from the bow
- The anchors provide the forward restraint and set the heading
- Both cables must be tended as the ship works



Aft

Surrounding buoys

- The stern is secured to the buoys surrounding the berth
- Lines are led from the poop to each buoy in turn
- The buoy pattern fixes the ship in position over the submarine line



Limits

Weather dependent

- Suitable for good to moderate weather and sea conditions
- Deteriorating weather is a reason to review the operation
- The arrangement gives no ability to swing to wind or current

Mooring points at a buoy berth

At a multi buoy berth the ship is held by a combination of her own anchors and lines to buoys in four groups.



Ship's anchors

Two anchors laid from the bow provide the forward restraint.



Quick release hooks

The mooring buoys are fitted with quick release hooks so lines can be let go from the buoy.



Beam mooring

Lines led out abeam to the buoys on either side hold the ship on the berth line.



Quarter mooring

Lines from the quarters to the after buoys restrain yaw.



Stern mooring

Lines from right aft to the stern buoys complete the pattern.



SECTION 02

Safe Mooring Practice

The risk of accident and injury during mooring and unmooring is higher than at any other time on board.

02

SECTION 2.1

Mooring stations carry the highest injury risk on board

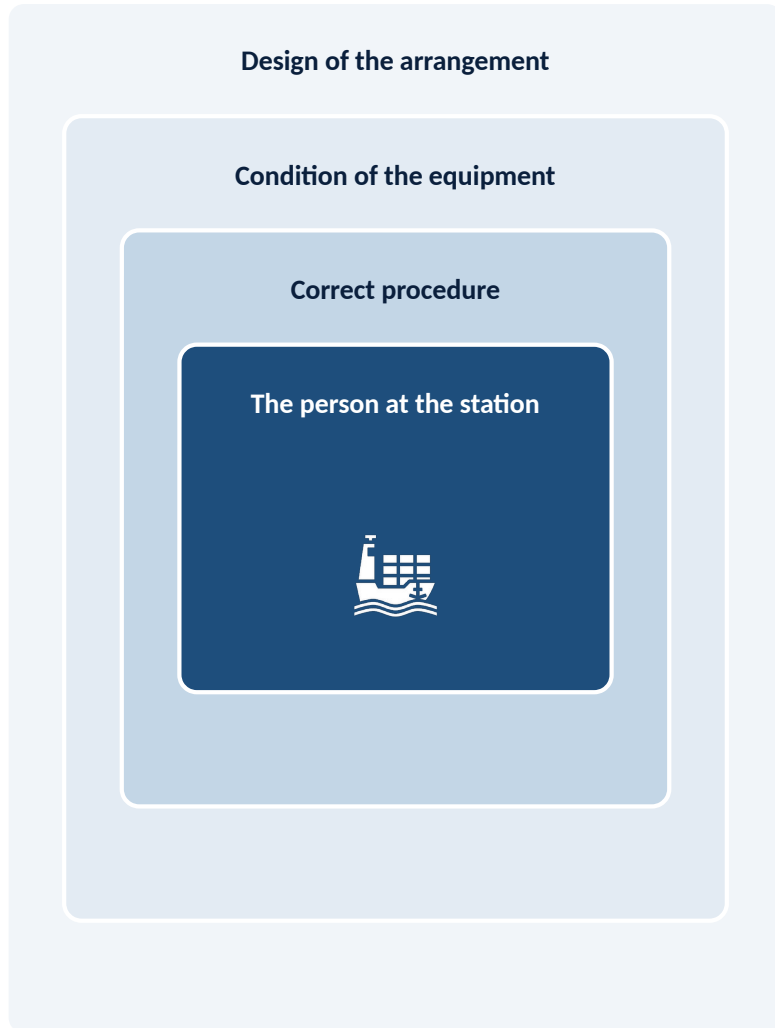
The risk of accidents and injuries during the mooring and unmooring manoeuvres of the ship is higher than at all other times. Nothing in this section is optional.

Rig it right

Stand clear

Watch the load

How the mooring system is protected, layer by layer



Design of the arrangement Enough lines, of the right material, led at the right angles, so that no line exceeds 55 per cent of its MBL under the design forces.



Condition of the equipment Winch brakes set to 60 per cent of the line MBL, one layer on the tension part, ropes and wires within their withdrawal criteria.



Correct procedure The right stopper, the right number of turns, the right sequence of brake, claw and clutch, and lines tended throughout the stay.



The person at the station Full PPE, a clear order understood, line of sight to the operator, and a position outside every snap-back zone.

Personal protective equipment at the station

The following is worn by every person at a mooring station, without exception.



Helmet, chin strap done up

A helmet with the chin strap fastened. An unfastened strap makes the helmet useless.



Safety shoes

Safety footwear that protects against dropped weights and gives grip on a wet deck.



Gloves

Gloves suitable for rope and wire handling.



Long-sleeved coveralls

Long-sleeved boiler suit covering arms and legs.



Eye protection for wires

Goggles are worn whenever a wire is being handled on the winch, because broken wires whip out.

General safety reminders at the station

✓ DO

- Wear a helmet with the chin strap tied, safety shoes, gloves and long-sleeved coveralls
- Wear goggles whenever a wire rope is being worked on the winch
- Make sure every order is understood before it is acted on
- Keep the person giving the order and the winch operator in sight of each other at all times
- Operate winch control levers with slow, deliberate movements

⊘ DON'T

- Never step into a bight of rope, or into the box formed by turns of a line on deck
- Never stand inside a coil or between a line and a fitting
- Never make sudden movements on a winch control lever
- Never work a line on an order you did not clearly hear and understand
- Never assume the operator can see you if you cannot see him

Handling ropes and wires on the winch

Most mooring station injuries happen at the drum end, and almost all of them are avoidable.

✓ DO

- Let a running line run — recover control with the winch, not with the body
- Keep well back from the warping drum when handling wire or synthetic rope
- Work the drum end only when a second person is available to feed the rope or take it away
- Have the line leader watch the whole of a tug line operation closely
- Keep tug lines slack while they are being made fast or let go

⊘ DON'T

- Never try to stop a running rope with your hand or your foot — serious injury will follow
- Never stand close to the rotating drum: rope can jump and trap your hands
- Never work a line on the drum end single-handed
- Never put load on a tug line while it is being made fast, or while the tug is standing by
- Avoid handling towing wires wherever it can be avoided

Orders and communication at the drum end

A rope under load is controlled by two people working as one. If the communication fails, the control fails.



Confirm every order

When a capstan or warping drum is used, make certain that the orders given have been understood before anything moves.



Keep line of sight

The person giving the order and the person operating the winch must be able to see each other throughout.



Never work alone at the drum

A rope on the drum end needs a second hand to feed it and to take it clear. One person cannot do both.



One voice at the station

Orders come from the officer in charge of the station, so that the operator is never given conflicting instructions.

A parting synthetic rope gives no warning at all

The greatest danger associated with synthetic ropes is the sudden release of stored energy when a rope under tension parts. Unlike wire, it gives no audible signal and may show no broken elements before it goes.

No warning

Full length of recoil

Stand outside the zone

Why synthetic rope behaves differently from wire



Wire rope

Gives warning

- Individual wires break progressively before the rope parts
- Audible signals are given as the rope approaches failure
- Broken elements are visible on inspection
- Low elasticity means less stored energy to release



Synthetic rope

Gives none

- Does not give audible signals of an impending problem
- May show no broken elements at all before it parts completely
- Stores a great deal of energy in its stretch
- Releases that energy instantly on failure — this is snap-back

Where a parted rope will actually go

The snap-back zone is not a fixed area painted on the deck; it depends on how the rope is led.



Straight lead: a straight recoil

If the line can travel back in a straight line then it will do so, striking anything or anybody in its path.



Around a lead: a wide arc

If a taut line is led around a lead it has the potential to whip round in a much bigger arc, well outside the original line of the rope.



Almost the full length

A parted rope will travel back almost its own length beyond its point of restraint.



Behind the drum as well

If the rope parts at some point behind the warping drum, the tail of the rope will fly beyond the drum.



The original direction is no guide

Where the rope passes over a pedestal roller, the direction of recoil may be nothing like the original direction of the rope.



Two points define the zone

The point of failure and the point of restraint together set the shape of the snap-back zone. Change the lead and you change the zone — so it must be reassessed for every berth.

Standing clear on the mooring deck

On the forecastle a headline, a breast line and a spring each generate their own snap-back zone at the same time.



Zones overlap

With several lines worked at once, the zones overlap and cover most of the working deck.



Stand outside, not beside

Position yourself outside the zone of every line under tension, not merely to one side of the rope.



Reassess as leads change

Each time a line is shifted to a different fairlead or roller, the zone moves with it.

Elongation and energy absorption of synthetic ropes

Percentages for the three common mooring fibres, new and used, with the energy absorption of new nylon taken as the 100 per cent reference.

Property	Nylon	Polyester	Polypropylene
Energy absorption — new rope, first use	100%	50%	43%
Elongation — new rope, first use	47%	36%	22%
Energy absorption — used rope	43%	20%	28%
Elongation — used rope	20%	15%	14%

The more energy a rope absorbs, the more energy it has to release if it parts.

What those percentages mean at the station



Nylon stores the most energy

New nylon absorbs twice the energy of new polyester and more than twice that of polypropylene. It therefore has the most energy to release, and the most violent snap-back.



Nylon also stretches the furthest

A new nylon rope elongates 47 per cent against 36 per cent for polyester and 22 per cent for polypropylene, so a parted nylon line travels furthest.



A used rope stretches much less

Elongation falls to 20, 15 and 14 per cent respectively once the rope has been in service.



Which changes how load is shared

A rope that stretches less picks up load sooner, so new and used lines in the same direction do not share load evenly.



And it is why materials are not mixed

The same reasoning is what rules out mixing wire and fibre, or new and worn rope, in the same direction.

Two ways of making fast on double bitts

Both methods are correct; the choice depends on the load and on how often the line will have to be adjusted.



Full turn, then figure of eight

For high load, infrequent adjustment

- One full turn is taken around the leading bitt before the figure of eight is started
- The classic figure of eight is then continued over both bitts
- The first turn takes much of the load off the crossed turns
- Preferred where the load on the line is high and the line will not need frequent adjustment



Direct figure of eight

For frequent adjustment

- The rope is taken directly in figures of eight, with no full turn first
- Accepted by most standards for load rating
- The line stays easy to control and easy to slack or heave
- Preferred where the line has to be adjusted constantly

Mixed mooring should never be the chosen method

Mixed mooring is the combination of wire and synthetic rope to secure the ship. It is a method that should never be preferred unless there is no alternative left.

Never by choice

Never in the same direction

Record the reason

If mixed mooring cannot be avoided

Where the ship is obliged to mix materials, the arrangement below limits the damage.

✓ DO

- Give the wire ropes as the breast lines and the springs
- Give the synthetic ropes as the head lines and stern lines
- Keep every line in one direction made of the same material
- Tend the lines continuously, because the two materials will not share load evenly
- Return to a single-material arrangement as soon as it is possible to do so

⊘ DON'T

- Never use two lines of different material together in the same direction
- Never accept a mixed arrangement as normal practice
- Never assume the wire and the synthetic line will take an equal share of the load
- Never leave a mixed arrangement unattended and untended
- Never mix materials to save time or effort

What mixed mooring looks like when it goes wrong

The photographs in this course show two real arrangements, both of which are unacceptable.



Wire and fibre side by side to the same bollard

Wires and synthetic lines led together in the same direction to the same shore point. The stiff wire takes the load first, then parts or overloads the fibre line.



Different materials on the same fairlead

Lines of different material sharing a fairlead and a pedestal roller. Their different stretch means only one line is actually working.



No load sharing at all

A wire stretches very little and a nylon line stretches a great deal. Together they cannot share load; one holds everything until it fails.



Failure spreads

When the first line goes, the whole of its load transfers instantly to the remaining lines, which then part in turn.

Stoppers — what they are for

A stopper temporarily secures a mooring line that would otherwise run free, while the line is being made fast.



A temporary hold

The stopper holds the line while it is transferred from the drum end to the bitts. It is not a permanent securing.



Synthetic fibre only, for fibre lines

A stopper for a synthetic rope must itself be made of synthetic fibre.



Short and easily handled

It should be easy to handle and as short as it can be made, so that it can be applied and cleared quickly.



Low stretch

It must be made of a material that does not stretch excessively, or it will not hold.



High melting point

Stoppers made up on board should be of a material with a high melting point — polyester or polyamide, for example.

The stopper strength rule

Where doubled rope is used as a stopper, its strength is set against the line it has to hold.

50%

Minimum strength of a doubled rope stopper, expressed as a proportion of the MBL of whichever mooring line is to be secured



Doubled rope stoppers — If double ropes are used as stoppers, they must have a strength equal to at least 50 per cent of the MBL of whichever rope is to be secured.



Never a fibre stopper on a wire — Synthetic rope must not be used to stopper a wire rope; it cannot obtain sufficient grip on the wire.



Check before use — Stoppers are working gear and wear like any other rope. Inspect before each use and renew when worn.

Rope stoppers — conventional and overlapping

Two methods of applying a rope stopper to a mooring line, each with a distinct point of care.



Conventional method

- The coils are laid on and the free end is held in the direction away from the standing end
- Disadvantage: the crossing point gets jammed and can become difficult to clear
- Clearing a jammed stopper under load is itself a hazard



Overlapping method

- The free end is held in the direction of the standing end
- Care: the stopper must be tightly overlapped, or the standing end will slip from under the coils
- Correctly applied, it clears more readily than the conventional method

Stoppering wire ropes

Wire ropes are stoppered in one of two ways only. A synthetic stopper is not one of them.



Carpenter stopper

A mechanical stopper clamped onto the wire, also known as an egg stopper or wire stopper. It grips the wire between jaws and is released by withdrawing the pin.



Chain stopper

A chain applied to the wire using a cow hitch. Simple, and effective on the wire surface where fibre would slip.

Use of the capstan and warping drum

- The rope is never made fast on the warping drum; instead three or four turns are taken and the operator controls the flow of the rope.
- Pulling force is adjusted by increasing or reducing the pressure held on the part of the rope coming off the drum.
- No more than three or four full turns of rope should be taken on the drum.
- Do not stand close to the rotating drum while it is turning.
- The distance between the operator controlling the rope and the drum should be at least 2.5 metres.
- Nylon rope should be avoided on the drum end wherever possible.
- High-gloss paint must not be applied to a capstan or warping drum, because the rope will slip on it.
- Under no circumstances is the mooring manoeuvre carried out with the mooring line on the capstan or warping drum; the equipment will be damaged.



The drum is not a bitt

A warping drum is a device for hauling a line in. The load is taken by the bitts or by the winch drum, never by turns on the warping end.

Two numbers for the drum end

3 – 4

Maximum full turns of rope on the warping drum

2.5 m

Minimum distance between the operator and the rotating drum



Control by friction, not by grip — The turns on the drum do the holding. The operator only varies the tension on the tail to control the rope.



No gloss paint — A painted, glossy drum surface will not hold the turns, and the rope will run.

The function of the winch brake

The brakes on the mooring winch and the anchor windlass are safety devices for the equipment they are fitted to.



The brake is meant to slip

Before the rope or chain on the drum parts under overload, the brake is intended to render and let it run.



Set to 60 per cent of MBL

The operating parameters of the mooring winch brake should be set so that it holds 60 per cent of the MBL of the line made fast. Above that load the brake must render.



Never over-tighten it

A brake tightened beyond its setting by the operator cannot do its job. Correct use matters as much as correct setting.



Better a rendered brake than a parted line

It is preferable for the winch to let the rope run than for the rope to break.



One failure leads to the next

If one line parts, the load on the remaining lines rises and they will part in turn.

Fire wires

Oil, gas and chemical tanker terminals may require fire wires to be rigged.



Their purpose

To allow tugs to tow the ship away from the berth without any assistance from the ship, in the event of a serious fire or explosion.



Terminal by terminal

Some terminals require fire wires today, some do not, and some require a different rigging arrangement.



Local rules govern

The recommended arrangement is given below, but local terminal rules must be followed where they differ.

Recommended fire wire arrangement

1

A wire of suitable size

The wire is to be in good condition and of a size suitable to the size of the tanker.

2

At least five turns on the bitts

The inboard end is made fast with at least five turns on the bitts.

3

Eye one to two metres above the water

The wire is led outboard through the chock and the eye is held one to two metres above the waterline.

4

Messenger line secured to the splice

A messenger line is secured to the splice and made fast inboard, so the eye can be recovered or paid out.



SECTION 03

Organising the Operation

Who is responsible for what, and the design criteria the mooring and anchoring systems are built to.

03

Responsibilities during the mooring operation



Pilot

Advises

- Advises the Master on how to organise the mooring operation and which line to give first
- Communicates with the tugs and with the shore mooring gang
- Holds detailed knowledge of the port and of the berth



Master

Overall responsibility

- Carries general responsibility for the whole mooring operation
- Must have gathered detailed information about the port and berth in advance
- Must know any specific port requirements on mooring, tug use, weather conditions and restrictions



Deck officer

Runs the station

- Takes instructions from the Master and manages the forward or after mooring station accordingly
- Instructs the station party on how each line is to be made fast
- Makes certain that the station party are following the safety precautions

What the Master must know before arrival

The information below is collected in advance, not asked for on the bridge during the approach.



The port and the berth

Detailed information on the port and on the specific berth the ship is going alongside.



Specific mooring requirements

Any particular requirement the port places on the mooring arrangement, including the number and type of lines.



Tug requirements

How many tugs are required, how they are to be made fast, and on which lines.



Weather limits and restrictions

The weather conditions and other restrictions that apply to berthing and to remaining alongside.



The plan briefed to the stations

The intended sequence of lines is passed to both stations so that everyone works to the same plan.

Mooring system design criteria

On a well-sited berth the greatest forces come from wind and current. Mooring systems are designed to the following general criteria.

Direction of the current	Wind	Current
From abeam	60 knots	0.75 knots
From ahead or astern	60 knots	3 knots
From intermediate directions	60 knots	2 knots

Wind speed increases with height above sea level, so freeboard is a critical factor for safe mooring.

Deciding how many lines to use

- On a well-sited berth the greatest forces acting on the ship come from wind and current.
- The number of lines is decided by evaluating the total maximum combined forces from all sources acting together.
- The calculation is made so that the resulting force on any line reaches no more than 55 per cent of its MBL.
- Wind speed increases with height above sea level, so freeboard is a critical factor in the assessment.
- When it comes to current, the under-keel clearance is the most critical factor determining its effect on the ship.
- If the under-keel clearance is less than the draught of the ship, the ship is deemed to have become an obstruction to the current.
- Taking ballast increases the current force but reduces the total force, because the wind effect is the larger of the two.
- Because freeboard and under-keel clearance change as cargo and ballast are worked, the assessment does not hold for the whole stay.



It is a combined assessment

No single force decides the arrangement. Wind, current, passing ships, swell and the ship's own changing condition are assessed together, and the arrangement is designed for the worst combination.

Design the arrangement to 55 per cent of the line MBL

The number of lines to be used is decided by evaluating the total maximum combined forces, and the calculation is made so that the force acting on the lines reaches no more than 55 per cent of their MBL.

55% MBL in service

60% brake setting

60 knots of wind

Freeboard, and why it decides the wind load



Wind speed rises with height

Wind speed increases with height above sea level, so the same nominal wind produces more force on a higher hull.



A light ship presents more area

A ship in ballast or part-loaded presents a much larger area to the wind than a loaded one.



Freeboard is a critical factor

For this reason freeboard is treated as a critical factor in assessing whether the mooring is safe.



It changes during the stay

Freeboard changes continuously as cargo and ballast are worked, so the mooring must be tended throughout.

Under-keel clearance and the force of the current

The effect of a 1 knot current on the same ship at four different under-keel clearances, expressed as a multiple of her draught.

Under-keel clearance	Force from a 1 knot current
5 × draught	50 tonnes
2 × draught	90 tonnes
0.5 × draught	150 tonnes
0.2 × draught	200 tonnes

Under-keel clearance is the most critical factor determining the effect of current on the ship.

Why a reduced under-keel clearance matters

When the under-keel clearance falls below the ship's draught, the ship becomes an obstruction to the flow.



What happens to the water

- If the under-keel clearance is less than the draught of the ship, the ship is deemed to have become an obstruction to the current
- The water either escapes around the bow or the stern
- Or it accelerates as it passes under the keel
- Either way the force on the hull increases sharply



The ballasting trade-off

- Increasing the draught by taking ballast reduces the under-keel clearance
- That increases the force of the current acting on the ship
- But it reduces the total force acting on the ship
- Because the effect of the wind is greater than the effect of the current

Anchor windlass design criteria — lifting power

Anchor windlasses are normally designed to the following criteria, where the chain diameter is in millimetres.

Chain grade	Lifting power
Grade 1 chain	$37.5 \times (\text{chain diameter})^2$
Grade 2 chain	$42.5 \times (\text{chain diameter})^2$
Grade 3 chain	$47.5 \times (\text{chain diameter})^2$

On these formulae the windlass is strong enough to lift three to four shackles of free chain plus the anchor, depending on the hawse pipe and the effectiveness of the devil's claw.

Maximum pull and how long it lasts

1.5 – 2 ×

Maximum anchor pull as a multiple of the nominal pull, depending on the classification society

2 min

Maximum duration for which the maximum pull may be applied



It is a short-term rating — Maximum pull is a rating for two minutes at most. It is not a working condition to be held.



Beyond it the motor suffers — Holding maximum pull for longer than the rating allows is how windlass motors are damaged.

Anchor windlass brake power criteria

The required brake holding power depends on whether a devil's claw is fitted.



With a devil's claw fitted

- The brake shall hold 45 per cent of the breaking load of the chain without slipping
- The devil's claw itself holds 80 per cent of the breaking load of the chain
- The two together carry the full load when the ship is anchored



With no devil's claw fitted

- The brake alone shall hold 80 per cent of the breaking load of the chain without slipping
- The whole holding duty falls on the brake
- Brake condition therefore becomes even more critical



SECTION 04

Winches and Windlasses

Mooring winch and anchor windlass types, their brakes, and the correct way to anchor and to recover.

04

How winches are classified

Mooring winches and anchor windlasses are described in four separate ways, and any machine is a combination of all four.



By type of control

Automatic tensioning or manual tensioning.



By drive type

Steam, electric or hydraulic.



By number of drums

Single drum, double drum or triple drum.



By drum and brake type

Split or undivided drum; manually applied or spring applied brake.

Automatic and manual tensioning



Automatic tensioning

- If the load on the line falls below a preset value, the winch automatically heaves the line in
- If the load rises above a preset value, the winch automatically pays the line out
- The winch works continuously without an operator present



Manual tensioning

- Heaving in and paying out is carried out by the operator in every case
- The line is adjusted only when someone goes to the winch to do it
- Requires the mooring to be tended deliberately throughout the stay

Drive types

Winch drums are turned by a motor and reduction gear driven by steam, electricity or hydraulic pressure.



Steam

Simple, safe and robust, with ideal speed and pull characteristics. Used on steam turbine ships. The driving force is usually provided by two pistons on a common crankshaft.



Electric

Mostly preferred on container and dry cargo ships. Pole-changing types offer several fixed speeds; frequency-changing types allow the speed to be set as desired.



Hydraulic

Hydraulic motors are usually fed from a central power pack. High-pressure types work at 150 to 320 bar.

Hydraulic drive — what the operator will notice

In a hydraulic system, flow and pressure do two different jobs, and this explains the behaviour of the winch.



Flow sets the speed

Hydraulic flow determines the speed of the winch drum.



Pressure sets the pull

Hydraulic pressure determines the pulling force the winch can develop.



Several winches at once

When more than one winch is worked at the same time, the available flow is shared, so heaving speed falls.



But the pull is unchanged

Pressure is not shared in the same way, so the pulling force of each winch remains the same.

Drum arrangements and drum types



By number of drums

- Single drum
- Double drum
- Triple drum



Split drum

Tension and storage parts

- Divided into a tension part and a storage part
- The line on the tension part is always a single layer
- This gives a constant brake holding capacity and a constant pulling force
- A significant advantage over the undivided drum



Undivided drum

Mostly smaller ships

- Encountered mostly on smaller ships
- If the line is not spooled onto the drum properly before it is loaded, the buried turns are damaged when the load comes on
- Holding capacity varies with how many layers are on the drum

The split mooring drum in detail

The main feature of a split drum is to obtain maximum pulling and brake holding force independent of how many layers are left on the drum.



One layer does the work

The result is obtained by using the first layer on the drum after having made the rope fast to the pulley.



Divided by a flange

The drum is divided into two separate parts by a dividing flange with a slot in it.



Storage part

Normally the larger part is used for storage and is called the storage part.



Tension part

The smaller part is used for tensioning and is called the tension part.



Ratings are for the first layer

Normally the hoisting load and the brake holding load are specified on the first layer of the drum.

Winch brake types

The brake is the most important part of the mooring system: it holds the ship's end of the line and saves the line from parting under excessive load.



Manually applied band brake

- The brake is tightened by human power on the brake handle
- The holding force achieved depends on the person applying it
- No indication of the holding force actually applied



Spring applied band brake

- The brake is controlled by hydraulic power, without human effort
- May be fitted with a brake force indicator showing the holding power being applied
- Gives a repeatable, known setting

More than one layer on the tension part destroys the setting

If there is more rope than a single winding on the tension side of the drum, the ability of the winch to hold to its specified brake setting is reduced. The setting on the drawing is only true for one layer.

One layer only

Then the brake setting holds

Spool it properly

The anchor windlass brake



Band brake always fitted

Anchor windlasses are necessarily equipped with a band brake, which may be of the manual or the spring applied type.



Disc brake on some models

Some models also have a drop brake — hydraulically activated disc brakes fitted in addition to the band brake.



For dynamic control

The disc brake provides dynamic control while the cable is running out.



Much lower holding capacity

The maximum holding capacity of a disc brake is far lower than that of the band brake. Its purpose is not to stop the anchor and chain but to control their speed.

Marking the anchor chain

Chain markings are the only reliable indication of how much cable is out, whatever indicators the windlass may have.



Every shackle marked

The shackle lengths of the anchor chain must be marked with white paint and stainless steel wire seizing, even where the windlass has a chain counter.



Two shackles before the bitter end

It is recommended that the shackle two before the bitter end lug is marked in red, so that it is clear the end of the chain is approaching.

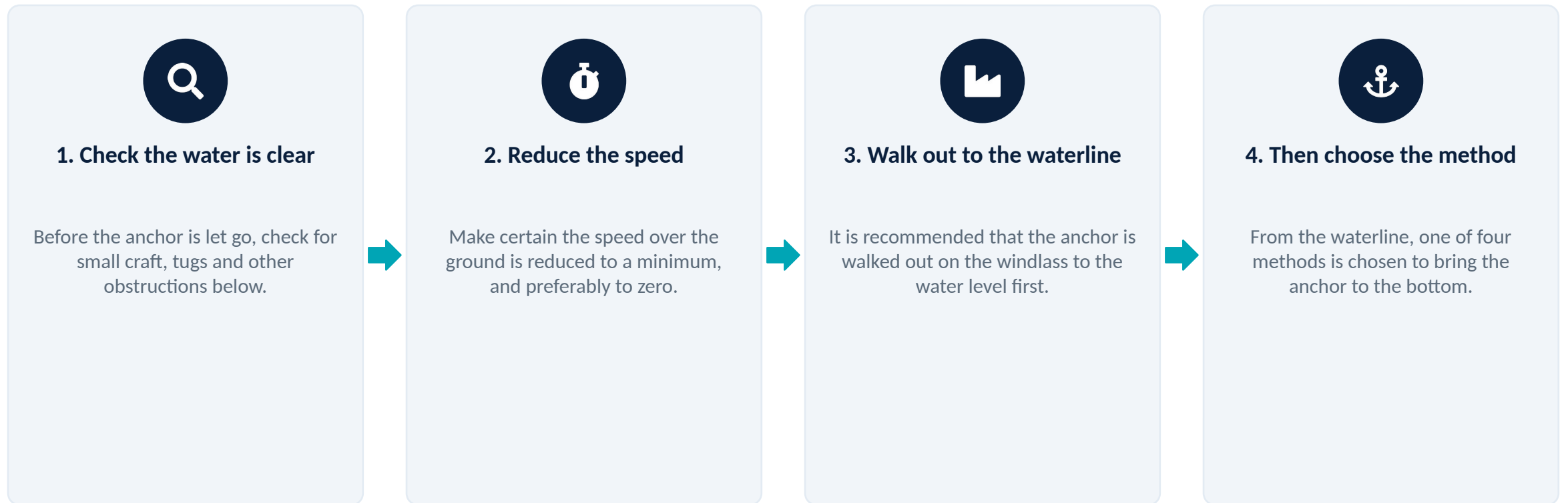


Read the chain, not the counter

Counters fail and drift. The painted and seized marks are what the officer on the forecastle actually reports.

Anchoring — the common opening sequence

The Anchoring Procedure section of the Company Manual must be complied with. Within it, the sequence below is recommended.



Four methods from the waterline down



Continue walking out

Read the maker's instructions first and learn to what depth the windlass may be used to walk the anchor out.



Let go on the brake

Tighten the brake, disengage the clutch, then release the brake in a controlled way. Snub with the brake to keep the anchor under control. The sharp line between open and closed makes a smooth run hard to achieve.



Automatic system

Where the ship has an automatic system, the chain length is entered into the system and the operation is left to it.



Control on the disc brake

Where a disc brake is fitted in addition to the band brake, tighten the band brake, disengage the clutch, tighten the disc brake fully, release the band brake completely, then start the run by slowly easing the disc brake to obtain the rate required.

Anchoring — the rules that protect the equipment

✓ DO

- Walk the anchor out where possible: it gives better control and extends brake lining life
- Check the water below for small craft and tugs before letting go
- Reduce the ship's speed to a minimum, and preferably to zero, before anchoring
- Mark the chain shackles, and report cable out from the marks
- On completion, tighten the brake fully, apply the devil's claw if fitted, then disengage the clutch

⊘ DON'T

- Never leave the clutch engaged while the ship lies at anchor
- Never let go without knowing the depth to which the windlass may be walked out
- Never let go with way still on the ship
- Never rely on a chain counter alone
- Never leave the brake partly applied and the devil's claw off

Why the clutch is disengaged at anchor

The order of operations at the end of anchoring is not arbitrary; it protects the windlass motor.



Brake fully applied first

When anchoring is complete, do not forget to tighten the brake fully before anything else.



Devil's claw on

Apply the devil's claw, or chain stopper, where one is fitted.



Then disengage the clutch

Only then is the clutch disengaged. It must remain disengaged for as long as the ship lies at anchor.



The reason

This prevents load reaching the windlass motor if the brake renders. If it did, the motor would be seriously damaged.

Recovering the anchor

- The chain must hang freely from the hawse pipe while the anchor is being recovered.
- If the chain enters the hawse pipe at an angle, stop heaving, tighten the brake and disengage the clutch.
- Resume heaving only when the chain has come to a vertical position.
- In bad weather, avoid recovering the anchor as far as possible by acting early rather than late.
- If the anchor must be recovered in bad weather, take care to avoid sudden loads on the windlass motor.
- A sudden load will cause a very serious windlass motor failure.
- The anchor must never be recovered in high speed mode.
- In high speed mode the pulling force of the windlass falls significantly, and if chain tension rises the risk of damage to the motor increases.



Act early in bad weather

The decision to weigh anchor in deteriorating weather is best taken before the swell builds. Once the ship is snubbing on her cable, every load on the windlass motor is a shock load.

The windlass motor is the weakest link in the anchoring system

Every rule about speed mode, chain angle, brake use and clutch position exists to keep load off that motor. If the motor goes, the ship has no ground tackle.

Low speed only

Chain up and down

Lever to neutral

If the windlass motor comes under high chain tension

With the hydraulic motor connected and very high chain tension, the operating lever must be kept in the neutral position.

✓ DO

- Keep the operating lever in the neutral position
- Apply the cable lifter brake
- Disengage the clutch between the anchor winch and the cable lifter unit
- Close the chain stopper bar if the situation becomes critical
- Remember that static friction lets the motor hold more load stopped than turning

⊘ DON'T

- Do not take the control lever to heave or to walk out under high chain tension
- Do not attempt to heave through a rising tension and hope it eases
- Do not leave the clutch engaged once the tension is recognised
- Do not delay closing the chain stopper if the situation is deteriorating
- Do not work the windlass in high speed mode at any time during anchor handling

Chain angle and the load on the motor

The angle between the chain and the vertical changes throughout recovery, and so does the risk to the windlass.



At the start: a large angle

When heaving begins the angle between the chain and the vertical axis is large. Vertical movement of the ship induces only small extra tension in the chain.



After about half the cable

After heaving in about half the chain length the angle has moved in, and from there it varies from about 45 degrees to 90 degrees.



The dangerous sector

In this sector, with the anchor still embedded, any vertical movement of the vessel may induce very high tension in the chain.



The motor over-speeds

That tension can cause the hydraulic motor to render at high speed, far beyond its maximum permissible speed, and cause severe damage.



The safety valve cannot cope

When that happens the oil flow will be more than ten times the capacity of the safety valve, which will not be able to bleed it off.

The windlass, the anchor winch and the cable lifter

A windlass consists of a driving unit called the anchor winch, and a chain handling unit called the cable lifter unit.



With high speed interlock

- It is impossible to run the anchor winch in high speed mode while it is engaged to the cable lifter unit
- The interlock removes the possibility of operator error
- A sign plate on the winch shows which arrangement is fitted



Without high speed interlock

- It is possible to run the anchor winch in high speed mode even when engaged to the cable lifter unit
- Always operate the cable lifter unit in low speed
- General rule: never carry out anchor handling with the winch in high speed mode
- The pulling force is reduced and, if chain tension is high, the risk of severe damage to the hydraulic motor is high

Windlass strength against chain strength — worked example

For a 100 mm chain with a minimum breaking load of 7,060 kN (720 tonnes).

Item	Basis	Value
Chain MBL	100 mm chain	7,060 kN (720 t)
Chain stopper holding capacity	80% of chain MBL	5,648 kN (576 t)
Cable lifter brake holding load	45% of chain MBL	3,177 kN (324 t)
Cable lifter nominal pull	$46.6 \times (\text{chain dia})^2$	466 kN (47.5 t)
Cable lifter maximum pull	nominal $\times 1.5$	699 kN (71.2 t)
10% of chain MBL	the pulling force limit	706 kN (69.9 t)

For any cable lifter unit the maximum pulling force is less than 10 per cent of the chain MBL.

What high speed mode does to the pull

On a cable lifter without a high speed interlock it is fully possible to run the windlass in high speed mode. This is what it costs.

1.7 – 2.0

Factor by which both nominal and maximum pulling force are reduced in high speed mode

5 – 6%

Resulting pulling force as a proportion of the chain MBL, against about 10 per cent in low speed



On the worked example — Five to six per cent of 7,060 kN gives 353 to 424 kN, that is 36 to 43.2 tonnes, against 71.2 tonnes in low speed.



Hence the rule — Keep the anchor winch in low speed mode when heaving in the anchor.

Fouled or jammed anchor — the procedure

Where the chain has fouled on the bottom and cannot be heaved, this procedure protects the windlass motor and the clutch.

1

Tighten the brake, claw off

Tighten the brake handle. Do not apply the devil's claw. If the foul frees suddenly, the brake can render and absorb the shock.

2

Disengage the clutch

Take the clutch out so that no shock load can reach the windlass motor.

3

Use the ship, not the winch

Use the ship's engine and rudder to work the fouled anchor clear, or to clear the turn in the chain.

4

Resume heaving

Once the problem is resolved, engage again and continue heaving normally.

5

Secure on completion

In the last metres reduce the heaving speed. When the anchor is home, tighten the brake, apply the devil's claw, secure the chain stopper and finally disengage the clutch.

The chain stopper



What it is

The chain stopper is one of the items of equipment that holds the anchor chain, designed to take the loads that may arise while the ship is at anchor.



Applied after the brake

It is used when the ship is anchored, and it is lowered into place after the brake handle has been tightened.



It backs up the brake

Its purpose is to add safety to the brake, not to replace it.



The safety pin is not optional

After the chain stopper is lowered the safety pin must be fitted. Without it the stopper can jump under sudden overload and the chain will run.



Anchor home in the hawse

Where the anchor is in the hawse pipe, the chain stopper should be in place along with the devil's claw.



SECTION 05

Synthetic Fibre Ropes

The six fibre materials used at sea, how to tell them apart, how to handle them, and when to condemn them.

05

The six fibre materials

Six synthetic fibres are met on board, but only some of them are acceptable as mooring lines.



Polyester

The most abrasion-resistant mooring rope. Multifilament, strong, and unaffected by water.



Nylon

Stronger than polyester but less resistant to abrasion. Multifilament. Loses strength when wet.



Polypropylene

The most commonly used because it is cheap and easily obtained, although OCIMF does not recommend it.



Polyethylene

Low friction monofilament with a low melting point and a low MBL. It cannot be used as a mooring rope.



Carbon fibre and aramid

Very strong, flexible, light and expensive, with a very small elongation. They do not float and must be jacketed.



HMPE

High-modulus polyethylene: as strong as steel wire, light, and floats. State of the art and very expensive.

Polyester



Best abrasion resistance

Polyester is the most durable mooring rope against abrasion. It is a multifilament fibre.



Constructions supplied

Supplied as three or four strand, twisted and jacketed; or as eight or twelve strand, double braided.



The heaviest

Polyester is heavier than all the other synthetic fibre ropes.



Less used on cost

It is less often preferred simply because it is expensive.



Unaffected by water

It has very little internal friction, is very strong, and does not lose strength when wet.

Multifilament and monofilament construction

The fibres themselves are made up in two ways, and the difference decides how the rope behaves in service.



Multifilament core

The strand is built from a very large number of fine filaments. Polyester and nylon mooring ropes are multifilament. The construction is strong, flexible and resistant to abrasion, because wear is spread over many filaments.



Monofilament core

The strand is built from a small number of coarse filaments. Polypropylene for mooring use is monofilament or split film, and polyethylene is a low friction monofilament. Coarse filaments give lower abrasion resistance and a lower melting point.

Nylon and polypropylene



Nylon

Strong, but wet strength falls

- Stronger than polyester, but less resistant to abrasion
- A multifilament fibre
- Supplied as three or four strand twisted, or eight or twelve strand double braided
- Nylon absorbs water, so there is a 10 per cent loss of strength when wet



Polypropylene

Common, but not recommended

- Many types exist; it is the most common synthetic fibre rope in use because it is cheap and easy to obtain
- OCIMF does not recommend it
- Supplied as three, four or eight strand
- Those used as mooring lines should be monofilament or split film type
- The melting point is low, so it can melt under excessive friction

Polyethylene, carbon fibre and HMPE



Polyethylene

- A low friction monofilament
- Low melting point and low MBL
- It cannot be used as a mooring rope



Carbon fibre and aramid

- Very strong, flexible and light
- Very small elongation percentage
- Expensive
- They do not float in water
- They must be jacketed



HMPE

- As strong as steel wire rope
- Lightweight, and floats on water
- State of the art, and very expensive
- Supplied as three strand twisted and jacketed, or as twelve strand braided

The six fibres compared

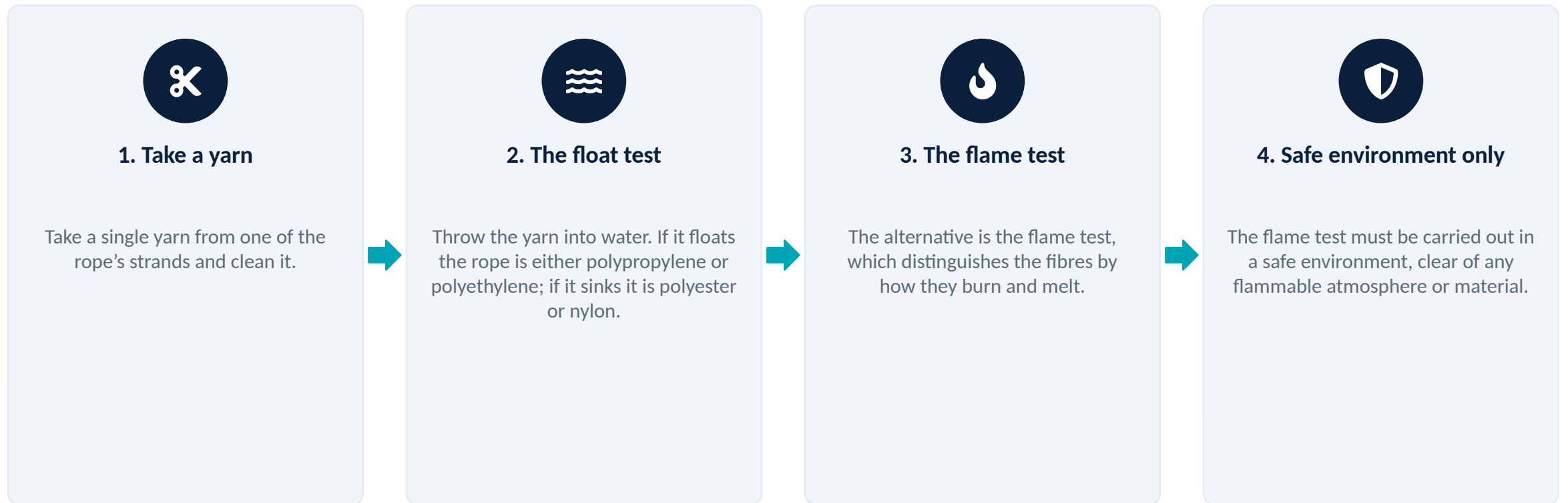
A summary of the properties given for each material in this section.

Material	Key property	Behaviour in water	Use as a mooring line
Polyester	Best abrasion resistance; heaviest; expensive	Sinks; no strength loss when wet	Yes
Nylon	Stronger than polyester; less abrasion resistant	Sinks; 10% strength loss when wet	Yes
Polypropylene	Cheap, easily obtained; low melting point	Floats	Common, but not recommended by OCIMF
Polyethylene	Low friction; low melting point; low MBL	Floats	No
Carbon fibre and aramid	Very strong, light, very low elongation; expensive	Does not float; must be jacketed	Specialised use only
HMPE	As strong as steel wire; light; very expensive	Floats	Yes

Fibre ropes are supplied as three or four strand twisted, or as eight or twelve strand braided constructions.

Determining what a fibre rope is made of

It is often difficult to tell visually what material a rope is made from. Two tests will separate the four common fibres.



Uncoiling and first use



Uncoiling a new rope

- A mooring rope is always unwound from the outside of the coil
- The coil should be seated on a turntable, or the reel rolled along the deck
- Taking the rope from the inside of the coil puts turns into it



First use of eight strand rope

- When eight strand rope is spooled onto the winch drum for the first time, the operation should be carried out under load
- That load should be approximately 10 per cent of the rope MBL
- This settles the construction and gives an even, firm first layer

Handling fibre ropes on deck

✓ DO

- Unwind from the outside of the coil, using a turntable or by rolling the reel
- Fit chafe protectors wherever the rope must bear against metal
- Keep drums, warping ends, rollers and swivels smooth and free of rust
- Keep the bearings of rollers and swivels well greased so that they turn freely
- Turn ropes end for end at set intervals to even out the wear

⊘ DON'T

- Do not run ropes over sharp edges
- Do not drag ropes along the deck, where they pick up abrasive residues and grit
- Do not lead ropes unnecessarily around bights, rollers, swivels or dolphins
- Do not allow a rope to rub against the deck at any point in its lead
- Do not leave a rusty or rough drum surface in service

Stowage — the three things that destroy a rope

A rope kept badly on deck can be ruined without ever having been under load.



Sunlight

Ropes should not be exposed to sunlight unnecessarily. If a rope is to be stowed on the open deck for a long period it must be covered with a tarpaulin.



Heat

No fibre rope should be exposed to excessive heat. Ropes must never be stacked near heaters in order to dry them.



Chemicals

No fibre rope should be exposed to chemicals, and acids and alkalis in particular. If contact is suspected, wash the rope immediately.

Inspecting a fibre rope

Rope inspection is done by hand, along the whole length, not by looking at it on the drum.



Every 30 centimetres, both hands

The person inspecting the rope examines it in both hands, working along the rope in 30 centimetre lengths.



External wear

Look for external abrasion and flattening of the outer yarns.



Cuts

Any cut into the strands is a reason to consider renewal.



Chemical damage

Softening, stiffening or discoloration from contact with chemicals.



Paint or heat damage

Glazing or fusing from friction heat, and paint that has stiffened the rope.

Tails and links — why wires need them

Wire mooring ropes give the most effective mooring system because of their low elasticity, but low elasticity brings its own problem.



The problem

The mooring system can be exposed to shock loads, especially at berths open to sea and swell, or at berths where ships pass close by.



Tonsberg and Mandal links

The problem is addressed at the end of the wire by fitting a Tonsberg type mooring link or a Mandal type fairlead shackle. Sizes are of the order of 60, 90, 120, 190 and 300 tonnes.



Fibre tails

The shock loading can be overcome by fitting a fibre rope tail — nylon, polypropylene or similar — at the end of the wire.



Watch the wire at the link

The parts of the wire rope attached to the links wear, and these parts must be checked at regular intervals.

Tail strength and renewal intervals

Because a synthetic tail wears out faster than the wire it is fitted to, its MBL is deliberately made higher.

+25%

Minimum MBL of a tail above the MBL of the wire it is attached to

+37%

Minimum MBL where the tail is nylon, because nylon loses strength when wet

18

Months after which OCIMF states that tails should be renewed



Inspect them more often — Tails should also be checked more frequently than the wires, or changed at regular intervals.



Check the wire at the link — The parts of the wire attached to the links wear, and must be checked at regular intervals.

What wear costs a fibre rope

Synthetic fibre ropes wear out faster than wire ropes, but it takes 12 to 18 months before the wear becomes noticeable.

–20%

Strength lost where one strand of the rope is broken

–50%

Strength lost where the outer yarns of all the strands are broken

–75%

Strength lost where the outer yarns and the second layer of yarns of all strands are broken



The wear is invisible at first — It takes between 12 and 18 months in service before the wear becomes noticeable to the eye.



Half strength is not half safe — A rope at half strength cannot hold the 55 per cent of MBL the arrangement was designed around.



SECTION 06

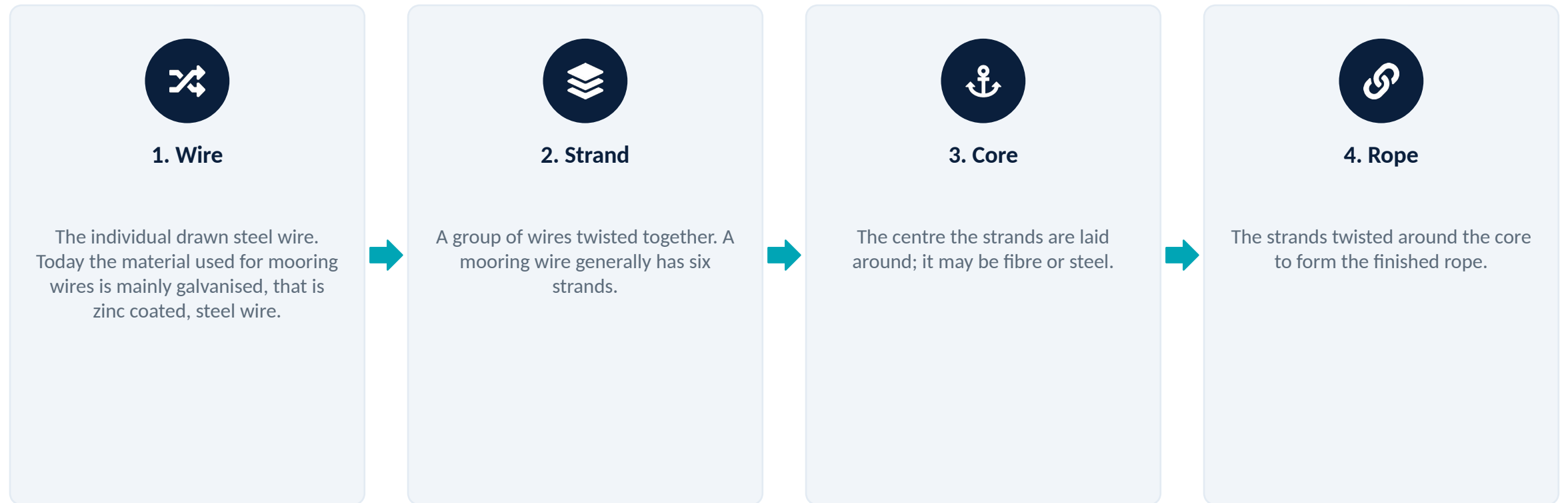
Steel Wire Ropes

How a mooring wire is built, how it is designated, how it is installed on the drum, and when it must come out of service.

06

How a wire rope is built up

Every element that makes up a wire rope is a spiral: wires are twisted into strands, and strands are twisted into the rope.



Recommended construction for mooring wires



Six strands

The recommended structure for wire ropes used for mooring is six strands.



31, 36 or 41 wires

Each strand consists of 31, 36 or 41 wires — for example 6 × 41 FC, IWRC.



Galvanised steel

The material used for this purpose today is mainly galvanised, zinc coated, steel wire.

Reading a wire rope designation

The construction shown in the course figure is designated 6 × 19 FC RH OL FSWR. Each element means the following.

Element	Meaning
6	Number of strands that make up the rope
19	Number of wires that make up each strand
FC	Fibre core
RH	Right hand lay
OL	Ordinary lay
FSWR	Flexible steel wire rope

Wire rope abbreviations — part one

Abbreviation	Description	Abbreviation	Description
FC	Fibre core	J	Jute (fibre)
FSWR	Flexible steel wire rope	LH	Left hand lay
FW	Filler wire	LL	Lang's lay
IWR	Independent wire rope	NR	Non-rotating
IWRC	Independent wire rope core	OL	Ordinary lay

Wire rope abbreviations — part two

Abbreviation	Description	Abbreviation	Description
RH	Right hand lay	TS	Triangular strand
S	Seale	W	Warrington
SF	Seale filler wire	WF	Warriflex
SW	Seale Warrington	WLL	Working load limit
SWL	Safe working load	WS	Warrington Seale

Fibre core against steel core

There is no exact rule for choosing between them, but there is a general practice for mooring wires.



Fibre core

Usually polypropylene

- Fibre cores are usually made of polypropylene
- The advantage is that the material does not absorb and retain moisture, which removes the cause of internal corrosion in the wire
- Not recommended where the wire is to be wound in several layers on the drum, because fibre cores are liable to crushing



Steel core

For hard service

- Can be used in very difficult operating conditions
- Suits a low safety factor, small drums and sheaves, and high operating speeds
- General practice for mooring wires: fibre core up to 38 mm diameter, and steel core above that

Hand and lay of a wire rope

The rope is right handed if the strands are twisted to the right, and left handed if they are twisted to the left.



Ordinary lay

Also called regular lay

- The winding direction of the wires forming the strands is different from the winding direction of the strands forming the rope
- Available as right hand regular lay and left hand regular lay
- The usual choice for mooring service



Lang's lay

- The winding direction of the wires forming the strands is the same as the winding direction of the strands forming the rope
- Available as right hand Lang's lay and left hand Lang's lay
- Presents a longer wearing surface to the drum

Matching the rope hand to the drum

The hand of the rope must match the drum, and there is a simple rule for working it out on the spot.

1

Face the drum

Stand facing the drum, with the flange to which the rope is to be attached on your side.

2

Thumb on the flange

Rest the thumb of your hand on the flange to which the rope is made fast, and point your index finger in the direction the rope is to wind.

3

If it is your right hand

A right handed rope is recommended. The pitch of the drum grooves is left handed.

4

If it is your left hand

A left handed rope is recommended. The pitch of the drum grooves is right handed.

Checking the diameter of a wire rope

The measured diameter is only meaningful if it is taken across the correct points.

✓ DO

- Measure across the widest points of the rope, that is over the crowns of two opposite strands
- Use calipers with jaws wide enough to bridge two strands
- Take the measurement at several points along the rope
- Record the measurement so that later reductions can be compared against it
- Compare the reading against the nominal diameter of the rope as supplied

⊘ DON'T

- Do not measure across the flats, that is into the valleys between strands
- Do not accept a single measurement as representative of the whole rope
- Do not measure over a damaged, kinked or flattened section and treat it as the diameter
- Do not use worn or sprung calipers
- Do not confuse a measuring error with a genuine diameter reduction

Taking the wire off the reel and onto the drum

Most wire ropes are damaged before they ever take load, during installation.



Discharge from the reel correctly

The reel is mounted so that it turns, or rolled along the deck. The wire is never pulled off the side of a stationary reel.



No turns induced

Pulling the wire off a stationary reel puts a turn into the rope for every coil taken off, and that is what causes birdcaging later.



Lead in line with the drum

The wire is led onto the drum in the correct plane. A lead across the drum spools the wire unevenly and crushes the lower layers.



Spool under tension

The first layer must be spooled tight and even, because it is the layer the brake rating is based on.

Winding onto the drum, and the fixing point



Right

Reel and drum aligned

- The reel is set up so that the wire comes off the top of the reel and onto the top of the drum, or bottom to bottom
- The wire leaves the reel and enters the drum in the same plane
- No reverse bend is put into the rope



Wrong

Reverse bend

- Wire taken off the top of the reel and led onto the bottom of the drum, or the reverse
- This puts a reverse bend into the rope as it is installed
- The rope is damaged before it has ever taken load



Fixing point on the drum

- The wire is made fast at the correct fixing point on the drum flange
- On a split drum this is on the tension side, so that the working layer is the first layer
- The fixing must be secure before any load is applied

Greasing the wire in service

A mooring wire is greased in place, on the ship, as part of routine work.



Grease bath on the lead

The wire is run slowly through a grease bath or trough placed in the lead between the reel and the drum, so that grease is worked into the rope as it passes.



Run it slowly

The wire is run slowly enough for the grease to reach between the strands, not merely to coat the outside.



Why it matters

The grease keeps water out of the rope, prevents internal corrosion, and reduces friction between the wires as the rope flexes.

Wire rope inspection — what to look for

Wire ropes wear in service and must be inspected at set intervals. Five defects are looked for in particular.



Birdcaging

The strands lift away from the core into a cage. It occurs after the rope has been unwound against its direction of lay. The rope is taken out of service directly.



Diameter reduction

A local reduction is usually caused by a break in the core. Reductions greater than 7 per cent condemn the rope.



Mechanical damage and wear

Crushing, flattening and abraded crowns from working over rollers, fairleads and the drum.



Kinks

A kink forms where a loop in the rope is loaded without the cross being taken out first. Serious loss of strength results.



Bends

Permanent bends caused by external influence. In most cases a bend puts the rope out of service.



Broken wires

Counted over a set length; the criteria are given on the next slide.

Diameter reduction and broken wire criteria

Three numerical criteria decide whether a wire rope stays in service.

7%

Diameter reduction above which the rope is taken out of service

10%

Broken wires, as a proportion of the total wires in the strands, over a length of 30 rope diameters

3

Broken wires in the termination, or on a single strand anywhere, that condemn the rope



Where to look for reduction — Places close to the terminations should be checked particularly carefully for diameter reduction.



How to count broken wires — If the total number of broken wires within a length equal to 30 times the rope diameter exceeds 10 per cent of the total number of wires making up the strands, the rope is taken out of service.



Three is the limit at the ends — Three or more broken wires in the end termination, or three or more on a single strand anywhere on the rope, are each a reason for withdrawal.

Wire rope — in service or out of service

A summary of the withdrawal criteria for a mooring wire.



MAY STAY IN SERVICE

- Diameter reduction within 7 per cent of the recorded original
- Broken wires below 10 per cent of the total, counted over 30 rope diameters
- Fewer than three broken wires at the termination and on any one strand
- Surface wear and light crown abrasion with the construction intact
- Rope well greased, round, and free of kinks and bends



TAKE OUT OF SERVICE

- Any birdcaging — the rope comes out of service directly
- Diameter reduction of more than 7 per cent
- More than 10 per cent broken wires over 30 rope diameters
- Three or more broken wires in the termination, or on a single strand anywhere
- A kink, or a bend caused by external influence



SECTION 07

Maintenance

Planned maintenance of the mooring winch and windlass, the checks, the lubrication and the brake tests.

07

Why planned maintenance pays

The maker's instructions on maintenance of the equipment must be strictly followed and the required checks carried out at the stated intervals.



Equipment lasts longer

Periodic maintenance extends the working life of the equipment.



Fewer accidents

Fewer accidents occur, because failures happen less often and at less dangerous moments.



Real economy

Serious economy is achieved over the life of the machinery.



Problems found early

Serious developing problems are noticed early, before they grow into failures.

Mooring winch periodic checks

The maker’s instructions govern in every case. The general convention for the checks is as follows.

Interval	Check
Before every operation	No abnormal noise and no overheating from the bearings. No abnormal noise from the clutch.
Once a week	No oil leak, and no leakage from the grease seals.
Once in three months	Condition of the brake lining. Grease injected from all grease nipples. All couplings working properly. All bolts and nuts tight. Drum drain opened and the water drained off.
Once a year	Hydraulic valves set correctly. Oil samples taken to confirm that no foreign matter or water is mixed in.

Lubrication of the mooring winch

The maker's drawing identifies every lubrication point on the machine and states what goes into each one.



Grease points

Gear case roller bearings, gear case couplings, drum bearings, warping head couplings and brake case bushes are all greased from their nipples.



Oil circulation

The gear case runs on circulating oil, which is filled and checked to the level given on the maker's drawing.



Work to the drawing

Because the points differ between makes and models, the lubrication drawing for the machine on board is the only authority.



On the three-monthly check

Grease injection from all nipples is part of the three-monthly periodic check.

Hydraulic valve settings

It is not recommended to change the settings of hydraulic parts in a system that is working properly. The following are checked periodically.

Valve	What it does	When to check
Flow control valve	Adjusts the speed of the winch	Annually; adjust if the winch speed is slower or faster than it should be
Pressure control valve	Adjusts the pulling force of the winch	Annually; adjust if the oil pressure is lower or higher than it should be
Counter balance valve	Controls the winch during load reduction	Check and adjust if there are irregular movements during load reduction or a rope break
Pressure sequencing valve	Controls automatic speed selection during operation	Annually; adjust if necessary

Brake holding capacity tests

The brake holding capacity test is the only way to know that a winch brake will actually hold what the mooring arrangement assumes it holds.



Once a year

Brake holding capacity tests should be carried out once a year.



Each lining separately

The test is carried out for each brake lining separately, so that a weak lining cannot be hidden by a good one.



Tested with a jack and gauge

The test is made with a hydraulic jack and pressure gauge applied against the drum, reading the load at which the brake begins to render.



Recorded and compared

Results are recorded and compared with the design brake holding capacity, which is set at 60 per cent of the MBL of the line.



Rig it right, stand clear, tend the lines

Mooring injuries come from ropes and wires that were led, stoppered or stowed wrongly, and from people standing where a parted line will go. Both are within our control.



Full PPE at every station



Stand outside every snap-back zone



55% MBL design, 60% brake setting



Maintain to the maker's instructions